

Inventory_based_disaggregation.R

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disaggregation	<i>Disaggregate emissions to the pixel scale using ACES or Vulcan CO2 inventories</i>
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Description

This is a function utilized by ‘stationary_combustion’ and ‘NG_distribution’. It distributes emissions from a larger scale (state or county) to the pixel scale using gridded CO2 emissions.

Usage

```
disaggregation(  
  input_inventory,  
  totals,  
  agg_level,  
  NEI_input,  
  cover_all,  
  out_envir  
)
```

Arguments

input_inventory	SpatRaster. Either a Vulcan or ACES sectoral inventory. ACES is available at https://doi.org/10.3334/ORNLDAAAC/1943 and Vulcan is available at https://doi.org/10.3334/ORNLDAAAC/1741 . Annual mean files should be used for Vulcan. The hourly ACES file should be averaged across hours to create an annually averaged inventory. Code to do this on a linux-based HPC system is available as the script "Annualize_ACES_seawulf.R" and the accompanying batch script "Annualize_ACES.sh".
totals	Character vector. Various subsectors to run through the disaggregation process.
agg_level	Character. The scale of the data before disaggregation, solely used for naming.

NEI_input	SpatVector. Includes attributes for each spatial feature that detail the emissions at the current agg_level separated by totals.
cover_all	List of matrices. Each entry in the list is a separate state/county and the matrices are the output of the function <code>terra::extract(weights=T,exact=T,cells=T)</code> . These provide the fraction of each pixel contained within the state/county. This allows for better accounting for pixels that are on the borders of states/counties.
out_envir	Environment. Where to assign output data.

Details

This function is intended to be used by ‘stationary_combustion’ and ‘NG_distribution’. For each pixel it calculates the ratio of CO2 from that pixel to the state/county it is in. This is combined with the state/county total CH4 emissions to disaggregate the emissions to the pixel-scale. If there are no emissions for the state/county then the entire area is assigned an equal fraction of emissions. This is done using sectoral (residential, commercial, industrial, electric) CO2 inventories.

ACES is available at <https://doi.org/10.3334/ORNLDAC/1943> and Vulcan is available at <https://doi.org/10.3334/ORNLDAC/1741>.

See references [Vulcan](#) and, [ACES](#)

Value

Nothing is returned. The disaggregated methane data is assigned to the specified out_envir. This will be a SpatRaster list with one for each of the totals.

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Examples

```
aces_res <- rast(paste0(ACES_directory,"/Sectoral/",ACES_year,'_Annual_ACES_Residential.nc'))
res_totals <- c('mains_ER_total_res',
               'serv_ER_total_res',
               'MnR_ER_total_res',
               'meter_ER_total_res',
               'upset_ER_total_res',
               'post_meter_ER_total_res')
cover_all <- list(extract(aces_res,all_merge_LCC_domain,weights=T,exact=T,cells=T))
disaggregation(aces_res,
               res_totals,
               agg_level="state",
               NEI_input=all_merge_LCC_state,
               cover_all,
               out_envir=environment())
```

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